Darshana G

Shimoga, Karnataka

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Summary

MCA student specializing in Full-Stack Development, with strong foundations in Data Structures Algorithms, Object-Oriented Programming, and Software Engineering. Proficient in Java, React.js, and TypeScript, with hands-on experience designing RESTful APIs and working with relational and non-relational databases. Strong problem-solving skills with a focus on building scalable and reliable software solutions.

Education

PES University **Sep 2025 – Present**
Masters in Computer Applications (MCA), CGPA: 7.9/10 *Bangalore, Karnataka*

LBAS And SBC College **Sep 2022 – May 2025**
B.Sc. in Mathematics, CGPA: 8.3/10 *Sagar, Karnataka*

Govt Pre-University College **July 2020 – May 2022**
Pre-University, PCMB, Percentage: 92% *Sagar, Karnataka*

Projects

LearnPulse : Reliability Adaptive Assessment & Learning Platform*RAG, Python, Next.js, open source LLM, ChromaDB, Typescript*

- Engineered a **Retrieval-Augmented Generation** (RAG) pipeline with ChromaDB vector search and a local LLM to generate **source-grounded** MCQs, using a groundedness gate and automated validation to reject hallucinated or malformed questions.
- Built an adaptive testing engine on **Item Response Theory (2PL model)** that estimates student ability and item difficulty/discrimination from response data, and selects each next question at the learner's level.
- Delivered role-based teacher and student dashboards with a review-and-publish workflow that turns approved AI-generated quizzes into a reusable question bank, with mastery and submission analytics.

ESG Sustainability Assessment System *Python, Javascript, Scikit-Learn, Pandas, RandomForest, plotly*

- Developed ESG Trust Score prediction system on 722 global companies across 47 sectors, achieving **96.6% R^2 accuracy** using **Random Forest Regressor with hyperparameter tuning**.
- Engineered a complete **data pipeline** including null imputation, ordinal encoding, IQR-based outlier capping, and a custom **0–100 trust score** normalization with bias correction.
- Computed 4 domain-specific features — Risk Level, Performance Level, Sector Average Difference & Sector Rank Percentile.

DataLens – Data Visualization & Analytics System *FastAPI, React.js, SQLite, DuckDB, Groq AI*

- Built a full-stack platform enabling **natural language querying** of CSV/Excel/JSON files, deployed on Vercel.
- Engineered an **NL-to-SQL pipeline** for real-time queries on datasets up to **100K rows**.
- Built an **LLM-driven** insight engine that generates context-aware insights, 8–12 analytical observations, and follow-up questions.

Technical Skills

Core Computer Science:	Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Computer Networks.
Programming Languages:	Java, Python, Javascript, TypeScript, SQL
Frameworks & Libraries:	React.js, Next.js, FastAPI, Node.js, REST APIs
AIML:	Pandas, Numpy, Scikit-Learn, MatPlotLib, Retrieval-Augmented Generation(RAG).
Databases:	MySQL, PostgreSQL, MongoDB
Developer Tools:	GitHub, Vercel, VS Code, IntelliJ IDEA, Jupyter Notebook, Google collab, AWS.

Leadership & Activities

- Coordinated Physics exhibitions at Science Fest involving **200+** students across 100 teams, managing team participation and event flow.
- Managed spot allocation and scheduling for 100 teams while coordinating exhibitions and on-ground activities for **1,000+** visiting students from other universities.

Achievements

- Solved **300+** coding problems across platforms — **LeetCode Profile**